

Orthability and the Ground of Intelligibility (companion draft)

DRAFT - not peer reviewed.

Empirical validation not completed: no designed study has been run.

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Orthability and the Ground of Intelligibility: A Transcendental Argument from Rule-Governed Resolution

Status: complete philosophical DRAFT (R2, 2026-07-20). School-neutral analysis; not peer reviewed; no empirical validation; nothing here is established by the operational manuscript, and nothing here validates it. Claim types are labeled inline as analytic, transcendental, explanatory, teleological, metaphysical, revelational, or school-internal; no revelational or school-internal claim occurs in this paper (the creed-internal development is the separate Atharī companion). Supersedes nothing: the earlier claim-ladder assessment and outline are retained as supporting history (orthemic-modal-metaphysical-assessment.md, orthemic-companion-paper-outline.md); Thesis C's disposition is Decision 0008. Objection/reply detail: OBJECTIONS-AND-REPLIES.md. Sources: sourcing/COMPANION-SOURCING-LEDGER.md.

Abstract

Created systems — perceivers, diagnosticians, parsers, validators — resolve concrete cases into determinate placements under governing standards, and their episodes are genuinely sound or unsound. This paper asks what that plain fact presupposes. It develops (A) a two-axis analysis separating mode of instantiation (concrete vs objective) from normative status (sound vs unsound), for both first-order placements and their governing standards; (B) an account of orthing faculties, first-order and higher-order, embodied and apprehended, engaging reliabilism and proper functionalism; (C) a transcendental argument from orthability — the antecedent determinability, differentiability, and fitness-for-lawful-resolution of what is — to an underived ground of intelligibility, carefully distinguished from modal-ontological and contingency arguments; (D) a reductio against the wholesale creation or self-assembly of orthability, with its strongest objection stated and adjudicated rather than dismissed; (E) a replacement of the earlier token-essential-modality thesis by an evaluability-precondition thesis; and (F) a conditional argument that the ground, if intelligent, is not intrinsically incapable of determinate disclosure. Rival accounts — modal primitivism, Platonism, structural realism, nominalism, selected-function naturalism, semantic naturalism, divine conceptualism, infinitism, deflationism, and generic classical theism — receive standing treatment, and the paper closes by stating exactly which steps each rival blocks and what remains unresolved. The argument is transcendental and internal to modal-normative discourse; it is not presented as a neutral proof compelling every rational reader.

1. The phenomenon, imported without its authority

The operational manuscript describes an architecture in which concrete occurrences (orthemata) are resolved into repeatable operational state-types (orthemes) relative to a declared analysis, by governed processes (orthing) whose episodes carry separable verdicts of result correctness and pathway adequacy. From that work this paper imports only the phenomenon, never its authority: rule-governed, evaluable resolution happens; some placements are correct and some are not; some procedures are truth-conducive and some merely lucky. (Analytic/empirical import; firewall: no engineering result supports any conclusion below — the manuscript's evidence stops at the existence and structure of the practice.)

Vocabulary note. "Orthability" names a relational candidate property: reality's being apt for orthing — determinable, differentiable, modally ordered, fit for lawful resolution — always relative to the resolving practice's standards. The word is a candidate term of the project's benchmark-gated vocabulary and does no argumentative work: every occurrence can be replaced by the phrase above.

2. Two axes: concrete/objective and sound/unsound

A diagnosis exists in the clinician; a parse exists in the parser. Call these concrete orthemes: locally instantiated determinations, real as psychological or computational tokens whatever their accuracy. Distinct from them stands what actually obtains: the objective profile — in the operational notation, $O(m; A)$, the state-types the case truly instantiates under the declared analysis. The concrete determination $\hat{p}$ is sound when it corresponds (or adequately approximates, at the analysis's tolerance) to the objective profile, unsound when it fails to. So the scheme is not "true ortheme vs false ortheme" but a product: mode of instantiation $\times$ normative status. A mistaken diagnosis is fully concrete and fully unsound; its being concrete does not make it sound. (Analytic.*)

The same two axes apply one level up. There is the concrete metaortheme — the standard or method a practice actually uses ("appearance suffices for placement here") — and the objectively fitting metaortheme — the standard appropriate to that class of case; and a local rule token is sound or unsound according as it corresponds to the fitting standard. A laboratory that validates compounds by color alone has a real, concrete, operative standard which is unsoundly related to the fitting one. (Analytic.)

Reconciliation with the operational core (Decision 0001). Analysis-relativity does not soften objectivity: once an analysis is declared, $O(m; A)$ is fixed by the occurrence and the analysis, and every actor can be wrong about it. What the two-axis scheme adds is the normative reading of that gap: created minds do not manufacture their epistemic targets; they produce local determinations that either conform or fail to conform to an antecedent objective order. Soundness is correspondence under the declared analysis — not identity of the local representation with reality in every respect, and not anyone's endorsement. (Analytic, given D1.*)

3. Faculties of orthing

3.1 First-order and higher-order

Vision, memory, inference, testimony-assessment, and diagnosis are, in this vocabulary, forms of orthing: each resolves concrete inputs into determinate placements. Abstracting, a first-order orthing faculty is a stable capacity F_O taking orthemmata to (usually sound) concrete placements. Above it operates a higher-order (metaorthing) faculty F_M that resolves questions about the resolving itself: is appearance sufficient here? is this inference form valid? is this witness reliable? is this rule fitted to this class of case? Its outputs are concrete metaorthemes. (Analytic taxonomy.)

A further distinction matters. A faculty's governing standards may be embodied — operative in its functioning without being represented (a child recognizes faces under constraints it cannot state) — or apprehended — explicitly grasped, formulated, and evaluated by reflective cognition. Sound method is typically embodied before it is propositionally apprehended, and apprehension is what makes correction, teaching, and audit possible. (Analytic/empirical.)

3.2 Reliabilism, proper function, and what they leave open

Process reliabilism (Goldman 1979) explains epistemic standing through the truth-conduciveness of belief-forming processes; it faces the generality problem (Conee & Feldman 1998): which of the many types a process token instantiates fixes the reference class? The operational framework's insistence on declared reference classes (RelSpec) is a governance answer, not a metaphysical one — it shows how a practice can discipline the choice, not that nature has made it. (Explanatory; concession recorded.)

Proper functionalism (Plantinga 1993) adds what bare reliabilism lacks: the notions of proper and improper functioning, a congenial environment, a design plan aimed at truth, objective success conditions, and the

possibility of malfunction. In the present vocabulary: a faculty is not merely de facto truth-tracking; it is ordered toward sound resolution — toward correct identification, valid inference, fitting evidential weighting, appropriate classification, correct practical handling, and the recognition and repair of defective method. Orthemology's contribution to this exchange is only to widen "truth-directed" to "sound-resolution-directed": the target is not only true propositions but sound placements and sound standards. (Explanatory.)

3.3 The teleological step, stated with its price

It is tempting to argue: reliable, properly functioning, truth-directed faculties point to a prior Knower who knows both the targets and the fitting means of apprehending them. This inference is not available from reliability alone. The naturalized alternative is serious: selected-effect accounts (Millikan 1984) ground function in selection history — a faculty's function is what its ancestors were selected for — and deliver malfunction, normativity of a kind, and truth-directedness of a kind, with no designer. To reach a Designer one must add at least two further premises and defend them:

- T1 (irreducible teleology): the truth- and soundness-directedness of cognitive faculties is not exhaustively reducible to selection-for-fitness — e.g., because selection explains usefulness and underdetermines aim at objective soundness as such, or because the normative surplus of "ought to have placed correctly" outruns "was selected because placings of this kind correlated with survival." (Teleological; contested.)
- T2 (causal adequacy): whatever grounds a genuine ordering of faculties to objective targets must itself be adequate to both — knowing the targets, knowing the fitting means, and capable of instituting the fit. (Metaphysical; a strengthened "principle of proportionate causality"; contested.)

If T1 and T2 hold, the design conclusion follows with content: the ground would need to know the objective orthemes, know the fitting metaorthemes, wisely fit faculties to targets, will their instantiation, and have power to create them — Knowledge, Wisdom, Will, Power, arising here from the proper-function architecture rather than being bolted on. If T1 fails, the selected-function account stands and this route stops at biology. The paper endorses the conditional and records both antecedent premises as contested. Plantinga's evolutionary argument against naturalism (2011) is a neighbor of T1, not a substitute for it. (Teleological/metaphysical, conditional.)

4. The transcendental argument from orthability

4.1 What kind of argument it is not

- Not a modal ontological argument. That argument runs from the possible necessary existence of God through a modal system to actuality ($\Diamond\Box G \rightarrow \Box G$; Oppy 2019 surveys). Here, by contrast, modality is partly the explanandum: the argument begins from actual determinacy, truth and error, valid and invalid inference, function and malfunction, fittingness and misfit, possibility and impossibility, successful and failed resolution — and asks what makes those distinctions real and correctly applicable. (Analytic contrast.)
- Not a standard contingency argument. The contingency argument uses the categories of possibility and necessity on beings; this argument asks why those categories — and their correct application — are available at all (Reichenbach 2022 for the contrast class). (Analytic contrast.)
- Not an argument from ignorance. No step has the form "we cannot explain X, therefore G"; every step claims a presupposition relation, which is either exhibited or fails. (Methodological.)
- Not a psychological argument from comprehensibility. The premise is not that humans find the world intelligible (a fact about us) but that evaluations of resolution episodes are true — that soundness and unsoundness are really instantiated. (Methodological.)

What it is: a transcendental argument in the standard sense (Stern 2019) — from an undisputed actuality to its

necessary preconditions — and it inherits the standard Stroud-type limitation (Stroud 1968): such arguments constrain what the discourse participant must grant, and a determined sceptic may retreat to denying the starting point or reinterpreting the preconditions as features of our conceptual scheme. This paper's conclusion is accordingly an internal-coherence result: given the reality of evaluable resolution, an underived ground is required; it is not a neutral proof against every retreat. (Transcendental; scope conceded.)

4.2 The chain, with each step labeled

1. Actual orthing. Evaluable resolution episodes occur. (Empirical/analytic launch point.)
2. Orthability. Their possibility presupposes that reality is antecedently determinable, differentiable, and fit for lawful resolution — else there is nothing to resolve and no difference between resolving and failing. (Transcendental.)
3. Determinate normative-modal structure. Orthability includes real distinctions of correctness/error, possibility/impossibility, identity/difference, and fittingness/misfit — the evaluability precondition C' of Decision 0008: every evaluable episode presupposes a determinate space of placements possible-but-not-made and of ways failure was possible. (Transcendental; replaces the retired token-essential-modality thesis.)
4. The self-assembly problem. This structure cannot coherently be explained as arising, by an intelligible process, from an absolutely unintelligible state (§5 below). (Transcendental reductio, with its scope limit stated there.)
5. An underived ground. Therefore not all orthability is derived/constructed/created; some locus of it is underived. (Metaphysical conclusion of 1–4, conditional on their readings.)
6. Anti-primitivist and anti-Platonist bridges. Against brute primitivism: primitivism is not an explanation but the declaration that none exists; it "survives" only at the price of explanatory arrears (§7.1). Against Platonism: a realm of causally inert abstracta grounds neither the applicability of the structure to concrete occurrences nor the fit of faculties to targets; inert exemplars do not select, institute, or enforce anything (§7.2). These are explanatory-superiority claims, not demonstrations. (Explanatory.)
7. The move to an intelligent ground. What must the underived locus be adequate to? Determinate distinctions, norms, and possibilities are the kind of thing intellects trade in; a ground adequate to intelligibility-as-such is most naturally characterized as knowing. This is the argument's second explicitly conditional joint (with T2-style causal adequacy doing the work): if grounding intelligibility requires adequacy to it, and adequacy to meanings and norms is intellectual, the ground is intelligent. A rival may stop at step 5 with an impersonal necessary structure; the paper marks that exit. (Metaphysical/explanatory; conditional.)
8. Contingency and Will. The actual pattern of instantiation — that these possibilities are realized rather than others — is contingent; a ground of both the necessary structure and its contingent instantiation must select among possibilities: Will. (Metaphysical; conditional on a selection-requires-selector premise, stated, not smuggled.)
9. Actualization and Power. Selection without efficacy institutes nothing; the ground must be capable of actualizing what it selects: Power. (Metaphysical; same conditionality.)
10. Ordered fittingness and Wisdom. The fit of faculties to targets and of standards to case-classes is an ordering of means to ends; adequacy to it is Wisdom. (Teleological; inherits §3.3's T1.)
11. Communicative orthability. Among the possibilities reality sustains is determinate representation and disclosure: signs can mean, orders of words can carry judgment, standards can be taught. §8 develops what this does and does not show about speech. (Analytic observation feeding §8.)

The chain's honest summary: steps 1–5 constitute the transcendental core; steps 6–7 are explanatory-superiority arguments; steps 8–10 are classical metaphysical extensions each flagged with its extra premise; step 11 is a bridge to a conditional capacity claim. Nothing later strengthens anything earlier.

5. The self-assembly reductio

5.1 The strongest formulation

Suppose all orthability is created (or assembled, or emergent-from-nothing-intelligible). Then orthability passes from non-instantiation to instantiation. But for that transition to be intelligible as a production of orthability, there must already be: a determinate distinction between orthability and its absence; a real possibility of its being instantiated; conditions under which the actualization succeeds or fails; identity across the proposed transition; and a relation between the cause and the appropriate result. Those are already elements of orthability — determinacy, modal possibility, fittingness, lawful transition, correct actualization. So the creation of all orthability presupposes orthability; hence not all orthability can be created. Equivalently: an account saying intelligibility "assembled itself" describes the assembly through an intelligible structure that must already obtain — it has not generated intelligibility from its absence but smuggled intelligibility into the machinery allegedly producing it. And the point is not "orthing occurs, therefore orthability is necessary" — actual things may be contingent; it is that the coming-to-be of orthability would itself already presuppose orthability. (Transcendental reductio.)

5.2 The strongest objection

The opponent has a principled reply with three prongs. (i) Deny the model of emergence. The reductio assumes that any origination of orthability is itself an orthing-like, rule-governed transition with success conditions. The naturalist may deny this: fundamental emergence, or a brute initial state, need not be a "production" satisfying intelligibility conditions — asking for the lawful conditions of the coming-to-be of lawfulness may be a category mistake. (ii) Accept brute modal structure. Primitivism (§7.1) simply takes the modal-normative structure as fundamental and uncreated without a ground; the reductio's conclusion — "not all orthability is created" — is one the primitivist happily grants while refusing the further move to a grounding being. (iii) Demote the metalanguage. That our account of the transition uses intelligible notions shows something about explanation, not about the ontology explained; an explanatory metalanguage need not be ontologically prior to its subject matter.

5.3 Adjudication

The reductio is decisive against a specific class of opponents and not universally conclusive. It defeats any account that models the origin of intelligibility as an intelligible, norm-governed construction — including most actual naturalistic construction stories (evolutionary, social-constructivist, conventionalist), which do describe rule-apt transitions and therefore presuppose what they would build. It does not defeat the opponent who bites the bullet of prong (i) or (ii): one may hold that there is no intelligible story of origination to be had (brutalism about the initial state) or that modal structure is primitive. Against those positions the argument's force is comparative, not demonstrative: they purchase consistency by declaring the central fact inexplicable, and §7 argues that is an explanatory cost, not a refutation. Prong (iii) is the weakest: the reductio's premise is not that our description is intelligible but that the transition described — a passage with identity, possibility, and success conditions — instantiates orthability; an opponent who strips those features has retreated to prong (i). (Adjudication; the "universally conclusive" claim is expressly not made.)

6. The evaluability thesis (Thesis C replaced)

The earlier assessment carried Thesis C: "every orthing token is essentially modal," supported by a quus-style rule-individuation argument and facing dispositionalist, deflationist, and minimal-implementation objections. Per Decision 0008 this paper argues from the weaker C': every evaluable token presupposes a determinate possibility space for its evaluation. C' is all the transcendental chain consumes (step 3), and the objections to C do not touch it: a dispositionalist about token modality still needs the distinction between success and possible

failure to be real for evaluations to be true; a deflationist who keeps evaluation-talk true has granted C' under another name, and one who abandons its truth has exited the starting point, paying the Stroudian toll of §4.1. The rule-individuation argument is retained in the assessment file as an optional strengthening, explicitly not load-bearing. (Analytic + decision record.)

7. Rival views

Full objection/reply development: OBJECTIONS-AND-REPLIES.md. Standing summary, with each rival's blocking point:

Rival	What it grants	Where it blocks the chain	Standing assessment
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Brute modal primitivism	Uncreated modal-normative structure (step 5!)	Step 7 (no intelligent ground; structure is bedrock) Consistent; explanatorily arrears-running – it re-labels the explanandum as primitive. Unrefuted.	
Platonism / necessary abstracta	Objective, necessary structure	Steps 6–7 (ground is inert abstracta) Faces the applicability/instantiation problem (why does the concrete *participate*?) and offers no account of faculty-target fit. Unrefuted as ontology; weak as explanation.	
Structural realism	Objective structure of the actual world	Steps 2–3 read modally (structure may be actual-world-only) Must still ground the modal force of evaluation (possible-but-not-made placements); tends to collapse into primitivism or Platonism at that point.	
Nominalism / conventionalism	Concrete particulars + our classifying practices	Step 2 (no antecedent determinability; we impose) Founders on error: if classification is imposition, unsoundness of an imposition needs a standard the view disallows; sophisticated versions re-import objective similarity, i.e., orthability.	
Selected-function naturalism	Real (etiological) function and malfunction	§3.3 T1 (teleology reduced to selection) The serious naturalist option. Answered only conditionally (T1); if T1 fails, the faculty route stops at biology – recorded, not hidden.	
Semantic naturalism	Naturalized content (information/teleosemantics)	Step 11's richer reading; §3.3 Same shape as selected-function debate, at the level of meaning; unresolved in general philosophy of mind, and this paper does not pretend otherwise.	
Divine conceptualism	Intelligent ground of abstracta/logic (Anderson & Welty 2011)	Nothing before step 8; a fellow traveler that identifies the structure with divine thoughts Nearest theistic neighbor; differences (attributes vs thoughts; the Atharī paper's refusal to make possibility-space an entity) are doctrinal, handled there.	
Infinitism	Each level grounded in a prior level, without end	Step 5 (well-foundedness of constitutive dependence) Blocked only by a no-infinite-regress-of-grounding premise; that premise is defended as the norm in grounding theory but is a premise. Recorded.	
Deflationism about normativity	The practice of evaluation-talk	Step 1 read realistically If evaluation-talk is kept truth-apt, C' follows; if not, the position abandons the datum. The retreat is available and priced (§4.1).	
Generic classical theism (no further identification)		Steps 1–10 Nothing; declines step 11's theological development Fully compatible terminus of this paper; the identification of the ground with the God of a particular revelation is *revelational/school-internal* and is made only in the Atharī companion.	

What remains unresolved, stated plainly: T1 against selected-function accounts; the selector premise at step 8; well-foundedness against infinitism; and the Stroudian retreat at the very start. The chain's conclusion therefore has the status: the best explanation available to one who takes evaluable resolution realistically, with the named exits open and priced. (Assessment.)

8. A ground capable of disclosure

Suppose the chain through step 7 (intelligent ground), with its conditionality acknowledged. What follows about speech? The argument, carefully bounded:

1. Orthability includes communicative orthability (step 11): determinate representation, fitting signification, ordered expression, reception, and correction are among the possibilities reality sustains — created speakers actually instantiate them. (Analytic/empirical.)
2. The ground of orthability is adequate to all of it — including the possibility-space of meaning and its fitting modes of expression (T2-style adequacy again). (Metaphysical; conditional.)
3. A knowing, willing, powerful ground adequate to meanings and their fitting expression is not intrinsically

mute: nothing in its nature excludes intentional determinate disclosure. To deny it the capacity for disclosure while granting it adequacy to communicative orthability would attribute to created speakers a perfection-relevant capacity whose ground lacks even the capacity — inverting the direction of adequacy. (Metaphysical, with the required perfection/causal-adequacy premise stated: capacities of effects are not surpassing their sustaining ground in kind.)

Boundary, stated as strongly as the claim itself: reason, so far, supports at most a capacity for intentional disclosure. Reason alone establishes nothing about whether the ground has actually spoken, in what words, with what letters or sound, on what occasions, or that any particular text is its speech. Every such particular is revelational and lies wholly outside this paper. (Boundary claim.)

9. Conclusion

From the reality of evaluable, rule-governed resolution, this paper argued transcendently to an underived ground of intelligibility; explanatorily against primitivist and Platonist restings; conditionally — with every extra premise flagged — to a knowing, willing, powerful, wise ground not intrinsically incapable of disclosure. The two-axis analysis and the faculty architecture stand on their own as philosophy of the operational framework, whatever becomes of the theology. The paper's negative results are as definite as its positive ones: reliability alone yields no designer; the *reductio* does not defeat brutalism or primitivism outright; the token-essential-modality thesis was not defended but replaced; and nothing here is a neutral proof. Where the argument's conditional joints resolve is, in the end, a matter for continued philosophy — and, beyond §8's boundary, not for philosophy at all.